ASSIGNMENT III: SQL TABLE MANIPULATION

Digital Submission Report & Documentation

Section 1: Project Overview

·Project Title: Assignment III - SQL Practical Exercise (Table Manipulation)

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·Business Idea: Pharmacy Management System

·Selected DBMS: MySQL 8.0+

Section 2: Database and Table Design

·Database Name: yournamestudentid_pharmacy_system

1. categories (Parent): Stores medicine category details.

2. suppliers (Parent): Stores medicine supplier details.

3.medicines (Child): Stores medicine details with Foreign Keys referencing categories and suppliers.

·ERD Diagram: The generated ERD is saved in the repository as erd.png.

Section 3: Database and Table Creation

```
CREATE DATABASE IF NOT EXISTS yourname_studentid_pharmacy_system;
USE yourname_studentid_pharmacy_system;

CREATE TABLE categories (
    category_id INT AUTO_INCREMENT PRIMARY KEY,
    category_name VARCHAR(100) NOT NULL UNIQUE,
    description TEXT
);

CREATE TABLE suppliers (
    supplier_id INT AUTO_INCREMENT PRIMARY KEY,
    supplier_name VARCHAR(150) NOT NULL,
    contact_phone VARCHAR(20) NOT NULL,
    email VARCHAR(100) UNIQUE
);

CREATE TABLE medicines (
    medicine_id INT AUTO_INCREMENT PRIMARY KEY,
    medicine_name VARCHAR(150) NOT NULL,
    category_id INT NOT NULL,
    supplier_id INT NOT NULL,
    unit_price DECIMAL(10, 2) NOT NULL CHECK (unit_price > 0),
    stock_quantity INT NOT NULL DEFAULT 0 CHECK (stock_quantity >= 0),
    expiry_date DATE NOT NULL,
    CONSTRAINT fk_category FOREIGN KEY (category_id) REFERENCES categories(category_id) ON
DELETE CASCADE,
    CONSTRAINT fk_supplier FOREIGN KEY (supplier_id) REFERENCES suppliers(supplier_id) ON
```

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```
DELETE CASCADE
);
```

Section 4: Initial Data Insertion

```
INSERT INTO categories (category_name, description) VALUES
('Antibiotics', 'Medicines used to treat bacterial infections'),
('Painkillers', 'Analgesic drugs used to relieve pain'),
('Vitamins', 'Dietary supplements and essential vitamins'),
('Antihistamines', 'Drugs used to treat allergies'),
('Cardiovascular', 'Medications related to heart and blood pressure');

INSERT INTO suppliers (supplier_name, contact_phone, email) VALUES
('PharmaCorp Ltd', '+250788111222', 'info@pharmacorp.com'),
('Global Health Supplies', '+250788333444', 'sales@globalhealth.com'),
('MediLife Rwanda', '+250788555666', 'contact@medilife.rw'),
('Afrimed Distributors', '+250788777888', 'orders@afrimed.com'),
('BioPharma Inc', '+250788999000', 'support@biopharma.com');

INSERT INTO medicines (medicine_name, category_id, supplier_id, unit_price, stock_quantity,
expiry_date) VALUES
('Amoxicillin 500mg', 1, 1, 1500.00, 100, '2027-06-30'),
('Paracetamol 500mg', 2, 2, 200.00, 500, '2028-01-15'),
('Vitamin C 1000mg', 3, 3, 5000.00, 50, '2026-12-31'),
('Cetirizine 10mg', 4, 4, 800.00, 200, '2027-09-20'),
('Amlodipine 5mg', 5, 5, 2500.00, 80, '2027-03-10');
```

Section 5: SQL Practical Tasks

5.1 SELECT Tasks

Q1.1: Basic Data Retrieval | Retrieve basic medicine details.

```
SELECT medicine_name, unit_price, stock_quantity FROM medicines;
```

·**Output & Explanation**: Displays product names, prices, and available stock.

Q1.2:Conditional Data Retrieval | Retrieve low stock items (<100).

```
SELECT medicine_name, unit_price, stock_quantity FROM medicines WHERE stock_quantity < 100;
```

·**Output & Explanation**: Filters medicines needing stock replenishment.

Q1.3: Sorted Data Retrieval | Sort medicines by price descending.

```
SELECT medicine_name, unit_price, expiry_date FROM medicines ORDER BY unit_price DESC;
```

·**Output & Explanation:** Lists medicines ordered from highest to lowest price.

5.2 JOIN Tasks

Q2.1:INNER JOIN | Match medicines with category names.

```
SELECT m.medicine_id, m.medicine_name, c.category_name, m.unit_price
FROM medicines m
INNER JOIN categories c ON m.category_id = c.category_id;
```

·**Output & Explanation:** Combines matching records from medicines and categories.

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Q2.2: LEFT JOIN | List all categories and associated medicines.

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```
SELECT c.category_name, m.medicine_name, m.stock_quantity
FROM categories c
LEFT JOIN medicines m ON c.category_id = m.category_id;
```

·**Output & Explanation:** Displays all categories even if no medicines are assigned.

Q2.3: Multi-Table JOIN | Generate comprehensive report.

```
SELECT m.medicine_name, c.category_name, s.supplier_name, s.contact_phone, m.stock_quantity
FROM medicines m
INNER JOIN categories c ON m.category_id = c.category_id
INNER JOIN suppliers s ON m.supplier_id = s.supplier_id;
```

·**Output & Explanation:** Combines 3 tables showing medicine, category, and supplier data.

5.3 UPDATE Tasks

Q3.1: Update One Record | Update price of Paracetamol (medicine_$i = 2$)

```
UPDATE medicines SET unit_price = 250.00 WHERE medicine_id = 2;
SELECT medicine_id, medicine_name, unit_price FROM medicines WHERE medicine_id = 2;
```

·**Output & Explanation**: Updates price of a single record and verifies change.

Q3.2: Update Multiple Records | Increase stock by 20 for low-stock items.

```
UPDATE medicines SET stock_quantity = stock_quantity + 20 WHERE stock_quantity < 100;
SELECT medicine_id, medicine_name, stock_quantity FROM medicines;
```

·**Output & Explanation**: Increases stock quantity for multiple qualifying items.

Q3.3: Conditional Update | Apply 10% discount on items expiring before 2027.

```
UPDATE medicines SET unit_price = unit_price * 0.90 WHERE expiry_date < '2027-01-01';
SELECT medicine_name, unit_price, expiry_date FROM medicines WHERE expiry_date < '2027-01-
01';
```

·**Output & Explanation:** Applies business discount rule to specific expiring items.

5.4 ALTER TABLE Tasks

Q4.1:**Rename Column |** Rename contact_phone to phone_number.

```
ALTER TABLE suppliers RENAME COLUMN contact_phone TO phone_number;
SELECT supplier_id, supplier_name, phone_number FROM suppliers;
```

·**Output & Explanation**: Renames column in suppliers table while preserving data.

Q4.2: Change Column Data Type | Change description data type to VARCHAR(255).

```
ALTER TABLE categories MODIFY COLUMN description VARCHAR(255);
DESCRIBE categories;
```

·**Output & Explanation**: Modifies column data type and verifies structural change.

Q4.3: Add Column | Add reorder_level column.

```
ALTER TABLE medicines ADD COLUMN reorder_level INT DEFAULT 10;
SELECT medicine_id, medicine_name, reorder_level FROM medicines;
```

·**Output & Explanation**: Adds new column with default value for inventory management.

5.5 INSERT Tasks

Q5.1: Insert **Single Record |** Insert one new category.

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```
INSERT INTO categories (category_name, description) VALUES ('Dermatology', 'Skincare and
topical ointments');
SELECT * FROM categories WHERE category_name = 'Dermatology';
```

·**Output & Explanation:** Adds a single record to categories table.

Q5.2: Insert Multiple Records | Insert two new suppliers.

```
INSERT INTO suppliers (supplier_name, phone_number, email) VALUES
('MedPharm Direct', '+250788123123', 'contact@medpharm.rw'),
('East Africa Pharma', '+250788456456', 'sales@eapharma.com');
SELECT * FROM suppliers WHERE supplier_id > 5;
```

·**Output & Explanation**: Inserts multiple records into suppliers table.

Q5.3: Insert Related Record |Insert medicine referencing new category and supplier.

```
INSERT INTO medicines (medicine_name, category_id, supplier_id, unit_price, stock_quantity,
expiry_date)
VALUES ('Hydrocortisone Cream 1%', 6, 6, 3200.00, 40, '2027-11-15');
SELECT m.medicine_name, c.category_name, s.supplier_name
FROM medicines m JOIN categories c ON m.category_id = c.category_id JOIN suppliers s ON
m.supplier_id = s.supplier_id
WHERE m.medicine_name = 'Hydrocortisone Cream 1%';
```

·**Output & Explanation**: Inserts child record maintaining valid foreign key constraints.

Section 6: Conclusion

Through this assignment, I successfully designed and populated a relational database using MySQL. I applied key database management concepts including primary/foreign key relationships, automated key generation, data retrieval using conditions and JOIN operations, safe data modification, structural schema alterations, and foreign key constraint validation.

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